

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY
2nd Semester of B.Pharm. Examination
University Theory Examination May 2017
MA141.1 Advanced Mathematics

Date: 17.05.2017, Wednesday Time: 10:00 a.m. to 01:00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section I are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat labeled sketches wherever necessary.
 6. Use of scientific calculator is allowed.
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SECTION – I

Q 1 Attempt all questions. Each question is of one mark. **20**

1. Fifth derivative of $y = e^{3x}$ is

[A] $y_5 = 3^5 e^{3x}$.

[B] $y_5 = 3^{-5} e^{3x}$.

[C] $y_5 = e^{2x+5}$.

[D] $y_5 = 2^{-5} e^{2x+5}$.

2. 15th derivative of $y = (7x - 65)^9$ is

[A] 1.

[B] -1.

[C] 0.

[D] 7.

3. Which of the following is true?

[A] Every Taylor series is Maclaurin series.

[B] Every Maclaurin series is Taylor series.

[C] Maclaurin series expansion around 1 is Taylor series.

[D] Maclaurin series expansion around 2 is Taylor series.

4. The order and degree of an ordinary differential equation $\frac{dy}{dx} - y = 0$ are

[A] order 1 and degree 1.

[B] order 2 and degree 1.

[C] order 1 and degree 2.

[D] order 2 and degree 2.

5. The first order ordinary differential equation $M(x, y)dx + N(x, y)dy = 0$ is exact if and only if
- [A] $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial x}$.
- [B] $\frac{\partial M}{\partial x} = \frac{\partial N}{\partial y}$.
- [C] $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.
- [D] $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial y}$.
6. The general form of linear differential equation is
- [A] $\frac{dy}{dx} + p(y)y = q(x, y)$.
- [B] $\frac{dy}{dx} + p(x, y)y = q(y)$.
- [C] $\frac{dy}{dx} + p(x)y^n = q(x), n > 1$.
- [D] $\frac{dy}{dx} + p(x)y = q(x)$.
7. Order of homogeneous function $f(x, y) = x^4 + 2x^3y + 5x^2y^2 + y^4$ is
- [A] 1.
- [B] 2.
- [C] 3.
- [D] 4.
8. Which of the following is a particular integral of $y'' + 4y' + 4y = 0$?
- [A] 0.
- [B] 1.
- [C] xe^{-2x} .
- [D] e^{-x} .
9. If roots of auxiliary equation are 1 and 2, then corresponding component (basis) solutions are
- [A] e^x & e^{2x} .
- [B] e^{-x} & e^{-2x} .
- [C] e^x & e^{-2x} .
- [D] e^{-2x} & e^{3x} .
10. Which of the following is true for the rate of change of number of bacteria(x) with respect to time(t) is inverse propositional to time?
- [A] $\frac{dx}{dt} = mt$.
- [B] $t \frac{dx}{dt} = m$.
- [C] $\frac{dx}{dt} = mt^2$.
- [D] $\frac{dx}{dt} = mt + 1$.

11. The rate at which bacteria multiply is proportional to the current number of bacteria. if x be the number of bacteria present at time t , then
- [A] $\frac{dx}{dt} = mx$.
- [B] $\frac{dx}{dt} = mx^2$.
- [C] $\frac{dx}{dt} = mx + kt$.
- [D] $\frac{dx}{dt} = e^{mx}$.
12. If T be the temperature of the body at time t and t_s is the temperature of surrounding, then Newton's law of cooling is given by
- [A] $\frac{dT}{dt} = k(T - t_s)$.
- [B] $\frac{dT}{dt} = kT + t_s$.
- [C] $\frac{dT}{dt} = kT$.
- [D] $\frac{dT}{dt} = k(t_s - T)$.
13. Solution of $x dx + y dy = 0$ is
- [A] $x + y = c$.
- [B] $x^2 + y = c$.
- [C] $x - y = c$.
- [D] $\frac{x^2}{2} + \frac{y^2}{2} = c$.
14. Maclaurin series expand function $f(x)$ at
- [A] $x = 0$.
- [B] $x = 1$.
- [C] $x = 2$.
- [D] $x = 3$.
15. According to Langrange's mean value theorem, if $f(x)$ is a continuous on $[a, b]$ and differential on (a, b) , then $\exists c \in (a, b) \ni$
- [A] $f(c) = 0$.
- [B] $f(c) = \frac{f(b) - f(a)}{b - a}$.
- [C] $f'(c) = \frac{f(b) - f(a)}{b - a}$.
- [D] $f(c) = \frac{f'(b) - f'(a)}{b - a}$.
16. 5th derivative of $y = (x + 3)^5$ is
- [A] 5!.
- [B] 6!.
- [C] 7!.
- [D] 8!.

17. 7th derivative of $y = e^{2x+3}$ is
 [A] $2^7 e^{2x+3}$.
 [B] $2^7 e^{2x-3}$.
 [C] $2^3 e^{2x+3}$.
 [D] $2^7 e^{2x}$.
18. If $y = \sin 2x$, then y_n is given by
 [A] $\sin 2x$.
 [B] $2^n \sin 2x$.
 [C] $2^n \sin\left(2x + \frac{n\pi}{2}\right)$.
 [D] $\cos 2x$.
19. The n^{th} derivative of $y = \frac{1}{ax+b}$ is
 [A] $\frac{(-1)^n n! a^n}{(ax+b)^{n+1}}$.
 [B] $\frac{(-1)^n n!}{(ax+b)^{n+1}}$.
 [C] $\frac{a^n}{(ax+b)^{n+1}}$.
 [D] $\frac{1}{(ax+b)^{n+1}}$.
20. The n^{th} derivative of $y = b^{ax}$ is
 [A] 0.
 [B] $(\log b)^n b^{ax}$.
 [C] $a^n (\log b)^n b^{ax}$.
 [D] $a^n b^{ax}$.

SECTION – II

Q 2 Attempt any **TWO** of the following.

- A** Bacteria multiply at the rate proportional to the number present. If the original number N doubles in three hours, then find required time so that the number of bacteria will be $4N$. 05
- B** Write statement of first shifting theorem and using it find Laplace transform of $t^2 e^{3t}$ and $e^t \cos 5t$. 05
- C** Verify Lagrange's mean value theorem for $f(x) = x^3 + 5$, $x \in [-1, 3]$. 05

Q 3 Attempt any **TWO** of the following.

- A** Verify Cauchy's mean value theorem for $f(x) = x^3$ and $g(x) = 2 - x$ where $x \in [0, 2]$. 05

- B** Identify the type of an ordinary differential equation $\frac{dy}{dx} + 4\frac{y}{x} = x^3$ and find 05
particular solution with help of initial condition $y(1) = 2$.
- C** Find general solution of $y'' + 2y' + 4y = 0$. 05

SECTION – III

Q 4 Attempt any **FOUR** of the following.

- A** Differential equation $\frac{ds}{dt} + 4s = e^t$ represents a relation between sugar(s) level in 05
blood and time(t) of a person. Solve the differential equation for sugar(s) level in
term of t.
- B** The population of a community is known to increase at a rate proportional to the 05
number of people present at a time t. If the population has doubled in 6 years,
how long it will take to triple?
- C** Evaluate $L(\sin 3t - t^3 + e^{-3t})$ and $L[(2t+5)^2 + 2e^{-t} + 5]$. 05
- D** Solve the differential equation $y'' + y = 0$ with initial condition $y(0) = 1$ and 05
 $y'(0) = 0$ using Laplace transform.
- E** Write statement of Leibnitz theorem for n^{th} derivative of product of two function 05
and using it find n^{th} derivative of $y = x^2 3^{5x}$.
- F** If rate of change of population is propotinal to time and at 1 hour rate is given by 05
100 person per hour, then find number of person at $t = 3$ hour and $t = 5$ hour where
at intial stage number of person is zero.

Q 5 Attempt any **FOUR** of the following.

- A** Find Maclaurin series of $\sin x$ and $\cos x$. 05
- B** Find Taylor series of function $f(x) = x^3 + 6x + 10$ around $x = 2$. Also show that 05
Maclaurin series is a particular case of Taylor series.
- C** Solve differential equation $\frac{dy}{dx} = \frac{x^3 + 2xy^2}{x^2 y}$. 05
- D** Find solution of $y'' + 6y' + 9y = 0$. 05
- E** Find solution of $y'' - y = e^x$. 05
- F** If rate of population(y-thousand) with respect to time(t-year) is represented by a 05
differential equation $\frac{dy}{dt} = t^3 y^{-1}$, then find population at $t = 3$ year where at intial
stage population is zero.